

ANURAG YADAV

Systems Integration & Architecture Engineer • ASEP

Infrastructure • Energy • Complex Projects

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PROFESSIONAL SUMMARY

Systems Integration Engineer with 8+ years of experience delivering complex commercial and industrial infrastructure projects across structural, MEP, and building services domains. Managed multi-project EPC portfolios exceeding \$40M, coordinating cross-disciplinary interfaces, sequencing dependencies, and operational constraints to improve schedule reliability, safety, and system performance.

Currently completing an M.Eng. in Systems Engineering at Cornell University to formalize systems-level decision-making, requirements analysis, and integration practices developed through real-world project execution, with a focus on infrastructure, energy, and large-scale socio-technical systems.

CORE COMPETENCIES

Systems & Integration Capabilities

- Cross-disciplinary systems integration (Structural, MEP, Building Services)
- Interface and dependency management
- Requirements interpretation and traceability (delivery-focused)
- System-level risk identification and mitigation
- Verification through commissioning and inspections

Infrastructure & Delivery

- EPC project execution
- Smart building and infrastructure systems
- Regulatory and safety compliance
- Stakeholder coordination across design, construction, and operations

Technical & Analytical Exposure

- Python (analysis, coursework projects)
- Systems modeling & MBSE (academic projects)
- Operations research & simulation (coursework)

EDUCATION

Master of Engineering in Systems Engineering

Expected May 2026

Cornell University Ithaca, NY

Relevant Coursework: Model-Based Systems Engineering, Digital Twins & Smart Infrastructure, Operations Research, Simulation Modeling, Advanced Project Management, System Architecture & Integration

Bachelor of Engineering in Civil Engineering

May 2016

Savitribai Phule Pune University Pune, India

Relevant Coursework: Structural Analysis, Structural Design, Project Management, Fluid Mechanics

CERTIFICATIONS

Associate Systems Engineering Professional (ASEP)

January 2026

International Council on Systems Engineering (INCOSE)

INCOSE Student Member

December 2025–Present

Attending INCOSE International Symposium 2026 (January 2026)

PROFESSIONAL EXPERIENCE

Avalon Developers (Commercial Real Estate Developer)

Pune, India

Project Manager – Infrastructure Systems

Nov 2021 – Jul 2025

- Directed \$25M+ portfolio of EPC infrastructure programs across 6 commercial and industrial projects, coordinating architectural, structural, MEP, and IoT system interfaces while maintaining 100% regulatory compliance and zero safety incidents across 500,000+ sq ft of developed space
- Pioneered digital transformation initiative deploying IoT sensor networks integrated with real-time scheduling and resource management platforms, achieving 20% reduction in operational downtime and \$800K in cost savings through predictive maintenance protocols
- Developed system-level risk assessments capturing interdependencies between structural integrity, MEP systems, and construction sequencing, reducing project delays by 18% and improving on-time delivery to 94%
- Led cross-functional teams of 15+ engineers spanning structural, mechanical, electrical, and civil disciplines across 6 concurrent projects, implementing iterative planning and coordination practices that accelerated delivery timelines by 22%
- Orchestrated stakeholder coordination with architects, regulatory agencies, and C-suite executives to ensure holistic system integration, resulting in 98% client satisfaction rating and 3 contract renewals totaling \$12M

Nyati Group (Residential & Commercial Developer)

Pune, India

Civil Engineer – Structural Systems

Nov 2018 – Oct 2021

- Led integration of structural, mechanical, and electrical subsystems across a \$15M commercial infrastructure portfolio spanning four high-rise developments, ensuring functional alignment, constructability, and full code compliance
- Improved workflow coordination and energy performance by aligning construction sequencing, resource allocation, and system interdependencies, contributing to a 15% increase in operational efficiency across 250,000 sq ft
- Developed and managed project schedules using scenario-based planning and iterative schedule reviews to reduce schedule variability by 28% and improve resource utilization by 19% across the project lifecycle
- Managed interfaces between architectural design, structural systems, and MEP installations on three LEED-targeted projects, preserving design intent while improving constructability and lifecycle cost outcomes
- Supported system-level commissioning, inspections, and performance verification activities, achieving full compliance with design and performance requirements and reducing post-occupancy defects by 31%

Manthan Construction India Pvt. Ltd. (Industrial & Commercial Construction)

Pune, India

Junior Engineer – Structural Systems

Jul 2016 – Sept 2018

- Managed Pre-Engineered Building (PEB) and Reinforced Cement Concrete (RCC) structural systems for 5 industrial and commercial projects totaling \$8M, coordinating vendor procurement and material quality control across 180,000 sq ft
- Optimized construction sequencing through detailed planning and coordination protocols, reducing material waste by 14% and improving schedule adherence to 89% through proactive interface management
- Ensured structural system integration with building services and architectural requirements, conducting constructability reviews that identified and resolved 45+ design conflicts pre-construction, saving estimated \$320K in rework costs

ACADEMIC & TECHNICAL PROJECTS

Smart Infrastructure Systems – CUSD Mobility Sustainable Bus Shelter | Cornell University

- Architected adaptive smart bus shelter system integrating IoT environmental sensors, real-time occupancy monitoring, solar energy harvesting (3kW capacity), and sustainable materials for deployment across Tompkins County transit network
- Applied Model-Based Systems Engineering (MBSE) using SysML in Cameo Systems Modeler to develop stakeholder requirements, system architecture, functional flow diagrams, and comprehensive validation protocols including FMEA analysis identifying 12 critical failure modes
- Executed multi-criteria trade studies balancing sustainability targets, user comfort metrics, operational efficiency, and 20-year lifecycle costs, demonstrating 35% energy savings vs conventional shelters and ROI within 8 years